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United States Patent	12413164
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Alva; Erick Falcon

Method for operating an electric machine, device, electric machine, and motor vehicle

Abstract

The invention relates to a method for operating an electric machine (9) comprising a rotatably mounted rotor (10) and at least one motor winding (13). In said method, a temperature load of the machine (9) is ascertained, and an electric motor current is applied to the motor winding (13) to generate a specified desired torque (N.sub.desired) according to the temperature load. According to the invention, when a temperature load is ascertained that exceeds a specified load threshold, a field strengthening mode of the machine (9) is activated, in which field-strengthening mode the motor winding (13) is energized in such a way that the motor current has a positive field-forming current component (Id).

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Appl. No.:	18/254719
Filed (or PCT Filed):	November 05, 2021
PCT No.:	PCT/EP2021/080723
PCT Pub. No.:	WO2022/117277
PCT Pub. Date:	June 09, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20240007034 A1	Jan. 04, 2024

Foreign Application Priority Data

DE

10 2020 215 352.3

Dec. 04, 2020

Publication Classification

Int. Cl.: H02P21/16 (20160101); B60L15/20 (20060101)

U.S. Cl.:

CPC H02P21/16 (20160201); B60L15/20 (20130101); B60L2240/421 (20130101); B60L2240/425 (20130101)

Field of Classification Search

CPC: H02P (21/16); B60L (15/20); B60L (2240/421); B60L (2240/425)

USPC: 318/473

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Background/Summary

BACKGROUND

- (1) The invention relates to a method for operating an electric machine comprising a rotatably mounted rotor and at least one motor winding, wherein a temperature load of the machine is ascertained, and wherein an electric motor current is applied to the motor winding to generate a specified desired torque according to the temperature load.
- (2) In addition, the invention relates to a device for operating an electrical machine having a control unit.
- (3) The invention also relates to an electric machine comprising such a device.
- (4) Further, the invention relates to a motor vehicle having such an electric machine.
- (5) An electric machine typically comprises a rotatably mounted rotor and at least one motor winding. In the operation of an electric machine, the motor winding is typically exposed to an electric motor current in order to generate a specified desired torque. The motor winding is thus energized or exposed to the motor current in such a way that a drive magnetic field is generated and the electric machine generates or provides the specified desired torque.
- (6) In operation, an electric machine heats up, wherein temperatures of greater than 230° C. can cause permanent magnets of a permanent magnet assembly of the electric machine to become demagnetized. An electric machine having at least partially demagnetized permanent magnets no longer provides the specified power or fails entirely when the permanent magnets are fully demagnetized. In order to protect the permanent magnets from demagnetization, it is known from the prior art to ascertain a temperature load of the electric machine and to take into account the temperature load when energizing the motor winding. For example, the publication WO 2019/096497 A1 discloses a method of operating an electric machine according to which the motor winding of the machine is energized at high temperatures in such a way that the magnetic stator flux of the machine is reduced compared to lower temperatures.

SUMMARY

- (7) The method according to the invention enjoys the advantage that the durability of the electric machine is increased. According to the invention, it is provided for this purpose that, when a temperature load is ascertained that exceeds a specified load threshold, a field strengthening mode of the machine is activated, wherein the motor winding in the field-strengthening mode is energized in such a way that the motor current has a positive field-forming current component (I_d). The electric motor current flowing through the motor winding generally consists of a torque-forming current component (I_q) and a field-forming current component (I_d). With respect to a rotor-fixed coordinate system, the field-forming current component flows parallel to the orientation of the magnetic field of the permanent magnet assembly. The torque-forming current component flows perpendicular to the orientation of the magnetic field of the permanent magnet assembly and corresponds to the torque generated by the machine. Depending on a working point of the machine, the torque-forming current component is positive, negative, or zero. The field-forming current component can also be generally positive, negative, or zero. According to the invention, the motor winding is energized when it is ascertained that the temperature load exceeds the load threshold in such a way that the motor current has a positive field-forming current component. Due to the positive field-forming current component, a field strengthening of the magnetic field of the permanent magnet assembly is achieved, from which the term “field-strengthening mode” is derived. Thus, due to the positive field-forming current component, a magnetic field is generated, which supports the correct alignment of micropoles of the permanent magnets of the permanent magnet assembly. In this respect, the positive field-forming current component prevents a demagnetization of the permanent magnet assembly of the machine or achieves a remagnetization of the permanent magnet assembly. Thus, a high temperature load of the electric machine is prevented from influencing the magnetization of the permanent magnets. This results in the increased durability of the electric machine achieved by the method according to the invention.

Preferably, in the field-strengthening mode, a desired current vector for the motor current is specified, which has a positive field-forming current portion, wherein the motor winding is then energized according to the specified desired current vector in such a way that the motor current has the positive field-forming current portion. Preferably, the rotor comprises the permanent magnet assembly. The permanent magnets are then connected to the rotor in a rotationally fixed manner. In this case, the motor winding is configured as a stator winding and is arranged distributed around the rotor.

(8) According to a preferred embodiment, it is provided that the motor winding is energized in the field-strengthening mode in such a way that the percentage share of the positive field-forming current portion of the motor current is at least 10%. This at least substantially prevents the demagnetization of the permanent magnets. Preferably, the proportion of the positive field-forming current component is at least 20%, particularly preferably at least 50%.

(9) According to a preferred embodiment, the motor winding is energized in the field-strengthening mode in such a way that the positive field-forming current component is as large as possible. As a result, the advantages described above with regard to the prevention of the demagnetization or achievement of the remagnetization are provided to a particularly high degree. The greatest possible level of the positive field-forming current component is in particular limited in that the torque-forming current component must have a certain level in order for the electric machine to generate the specified desired torque. Thus, according to the specified desired torque, the motor winding is set to a working point of the electric machine in which the specified desired torque is generated and the positive field-forming current component is as large as possible.

(10) According to a preferred embodiment, it is provided that, when a temperature load is ascertained that falls below the load threshold, the motor winding is energized in such a way that a torque-forming current component of at least substantially matches the motor current. The motor current is thus at least substantially free of a field-forming current component. As a result, an operation of the electric machine with particularly high efficiency is achieved. A positive field-forming current component is not necessary, because the load threshold is undershot.

(11) Preferably, the temperature load of the rotor is ascertained as the temperature load of the machine. This is advantageous in that the rotor or the permanent magnet assembly of the rotor is the part of the electric machine that is to be protected against temperature-related damage. By ascertaining the temperature load of the rotor, a particularly precise decision can be made as to whether the positive field-forming current component is advantageous or necessary in order to prevent the demagnetization or to achieve the remagnetization or whether the positive field-forming current component currently does not provide any advantage due to the low temperature load. Preferably, a temperature of a stator of the electric machine is sensed by means of a temperature sensor, and the temperature of the rotor is ascertained on a model basis according to the sensed temperature of the stator.

(12) Preferably, a temperature, in particular the temperature of the rotor, is ascertained as the temperature load. In this case, a threshold temperature is provided as the load threshold. For example, a temperature of between 210° C. and 250° C. is provided as the threshold temperature, particularly preferably a temperature of 230° C. Alternatively, a temperature integral of the temperature, in particular the temperature of the rotor, is preferably ascertained as the temperature load. In this case, a temperature integral threshold is then provided as the load threshold.

(13) According to a preferred embodiment, it is contemplated that a threshold speed is specified for the rotor and that the field-strengthening mode is activated only in the presence of an actual speed of the rotor that falls below the threshold speed. As a result of such a procedure, it is prevented that an undesirably high electrical voltage is induced into the motor winding due to the rotation of the rotor at high actual speeds of the rotor exceeding the threshold speed.

(14) According to a preferred embodiment, it is contemplated that a threshold torque is specified for the desired torque and that the field-strengthening mode is only activated in the presence of a

desired torque that is below the threshold torque. If the desired torque exceeds the threshold torque, the provision of the positive field-forming current component could influence the ability of the electric machine to generate the desired torque to the specified level. This is undesirable.

(15) The device according to the invention for operating an electric machine comprising a rotatably mounted rotor and at least one motor winding is characterized by a control unit specifically designed to perform the method according to the invention when used as intended. This, too, results in the aforementioned advantages.

(16) The electric machine according to the invention comprises a rotatably mounted rotor and at least one motor winding. This, too, results in the aforementioned advantages. Preferably, the electric machine is configured as a permanent-magnet synchronous machine. In this respect, the electric machine comprises a permanent magnet assembly. The permanent magnet assembly is preferably part of the rotor and is connected in this respect to the rotor in a rotationally fixed manner.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The invention will be explained in more detail in the following with reference to the drawings. The figures show:

(2) FIG. 1 a motor vehicle in a schematic view, and

(3) FIG. 2 a method for operating an electric machine of the motor vehicle.

DETAILED DESCRIPTION

(4) FIG. 1 shows a motor vehicle **1** in a schematic view. The motor vehicle **1** comprises a front wheel axle **2** with two front wheels **3** and **4** and a rear wheel axle **5** with two rear wheels **6** and **7**.

(5) The motor vehicle **1** also comprises a drive assembly **8** with an electric machine **9**, which is configured so as to drive at least one of the wheels **3**, **4**, **6**, and **7**. For example, the machine **9** is configured so as to drive the front wheels **3** and **4**, the rear wheels **6** and **7**, or both the front wheels **3** and **4** and the rear wheels **6** and **7**. Optionally, in addition to the machine **9**, the drive assembly **8** comprises at least one further electric machine and/or at least one internal combustion engine.

(6) The electric machine **9** comprises a rotor **10** that is rotatably supported in a housing of the machine **9**, which is not shown. The rotor **10** comprises a permanent magnet assembly **12** having at least one permanent magnet **11**. In the present case, several permanent magnets **11** are present. The permanent magnets **11** are fixedly connected to the rotor **10** and are arranged in a circumferential direction of the rotor **10**. For example, permanent magnets **11** are neodymium-iron-boron magnets **11**.

(7) The machine **9** also comprises a motor winding **13**. In the present case, the motor winding **13** is a stator winding **13** of a stator **23** of the electric machine **9**. In the present case, the stator winding **13** comprises three phases **14**, **15**, and **16**. The phases **14**, **15**, and **16** are arranged around the rotor **10** in such a way that the rotor **10** is rotatable by a suitable energization of the phases **14**, **15**, and **16**. In addition to the stator winding **13**, for example, the stator **23** comprises a carrier (not shown) for the stator winding **13**.

(8) The phases **14**, **15**, and **16** are electrically connected to an energy store **18** of the motor vehicle **1** by a power electronics **17** of the machine **9** comprising several switching elements.

(9) The electric machine **9** also comprises a device **19** for operating the machine **9**. The device **19** comprises a control unit **20**, which is configured so as to drive or switch the switching elements of the power electronics **17**.

(10) The motor vehicle **1** also comprises a speed sensor **21** associated with the rotor **10**. The speed sensor **21** is configured so as to sense the actual speed $N_{\text{sub.actual}}$ of the rotor **10**. The speed sensor **21** is communicatively connected to control unit **20** in order to provide the control unit **20**

with its sensor signal, i.e. the sensed actual speed $N_{\text{sub.actual}}$.

(11) The motor vehicle **1** also comprises a temperature sensor **22**. The temperature sensor **22** is associated with the stator **23** and is configured so as to sense the temperature of the stator **23**. Temperature sensor **22** is communicatively connected to control unit **20** in order to provide the control unit **20** with its sensor signal, i.e. the sensed temperature.

(12) An advantageous method for operating the electric machine **9** by means of the control unit **20** is described in the following with reference to FIG. 2.

(13) In a first step **S1**, the control unit **20** provides a desired torque $M_{\text{sub.desired}}$ that is to be generated by the machine **9**. For example, the control unit **20** ascertains the level of the specified desired torque $M_{\text{sub.desired}}$ according to an activation of an accelerator pedal by a driver of the motor vehicle **1**.

(14) In a second step **S2**, the temperature sensor **21** senses the temperature of the stator **23** and provides the sensed temperature to the control unit **20**.

(15) In a third step **S3**, the control unit **20** ascertains a temperature load of the electric machine **9** according to the temperature sensed by the temperature sensor **22**. In the present case, the control unit **20** ascertains the temperature of the rotor **10** as the temperature load of the electric machine **9**. The control unit **20** ascertains the temperature of the rotor **10**, preferably on a model basis, according to the sensed temperature of the stator **23**.

(16) In a fourth step **S4**, the speed sensor **21** senses the actual speed $N_{\text{sub.actual}}$ of the rotor **10** and provides the sensed actual speed $N_{\text{sub.actual}}$ to the control unit **20**.

(17) Steps **S1**, **S2**, **S3**, and **S4** are carried out continuously so that the desired torque $M_{\text{sub.desired}}$, the temperature load of the rotor **10**, and the actual speed $N_{\text{sub.actual}}$ of the rotor **10** are continuously ascertained or sensed.

(18) In a fifth step **S5**, the control unit **20** decides whether to operate the electric machine **9** in a standard mode or in a field-strengthening mode, i.e. in a field-strengthening operation. In step **S5**, the control unit **20** takes into account the specified desired torque $M_{\text{sub.desired}}$, the ascertained temperature load of the rotor **10**, and the sensed actual speed $N_{\text{sub.actual}}$ of the rotor **10**. Step **S5** is preferably also carried out continuously.

(19) The control unit **20** provides a threshold load for the temperature load, a threshold speed for the actual speed $N_{\text{sub.actual}}$, and a threshold torque for the desired torque $M_{\text{sub.desired}}$. In the present case, the control unit **20** provides a threshold temperature as a threshold load, particularly preferably a threshold temperature of 230° C. In step **S5**, the control unit **20** then compares the desired torque $M_{\text{sub.desired}}$ to the specified threshold torque, the ascertained temperature load to the specified threshold load, and the sensed actual speed $N_{\text{sub.actual}}$ to the specified threshold speed.

(20) If the comparison shows that the temperature load exceeds the threshold load, that the actual speed $N_{\text{sub.actual}}$ falls below the threshold speed, and that the desired torque $M_{\text{sub.desired}}$ falls below the threshold torque, then reference is made to a sixth step **S6**. In this step **S6**, the electric machine **9** is then operated in the field-strengthening mode. The control unit **20** then controls the switching elements of the power electronics **17** in such a way that an electric motor current flowing through the motor winding **13** has a torque-forming current component I_q and a positive field-forming current component I_d . With the torque-forming current component I_q , it is achieved that the electric machine **9** generates the specified desired torque $M_{\text{sub.desired}}$. With the positive field-forming current component I_d , it is achieved that a magnetic field is generated that is aligned parallel to the magnetic field of the permanent magnets **11**. This magnetic field prevents permanent magnets **11** from being demagnetized at a temperature load of the rotor **10** that exceeds the threshold load. In particular, it is achieved that portions of the permanent magnets **11** already demagnetized become remagnetized. In this respect, by operating the electric machine **9** in the field-strengthening mode, the durability of the permanent magnet assembly **12** and thus the durability of the electric machine **9** is increased.

(21) If, however, the comparison in step S5 results in the desired torque $M_{sub.desired}$ exceeding the threshold torque, the temperature load exceeding the threshold load, and/or the actual speed $N_{sub.actual}$ exceeding the threshold speed, then reference is made to a seventh step S7. In the seventh step S7, the electric machine **9** is then operated in the standard mode. The control unit **20** then controls the switching elements of the power electronics **17** in such a way that the electric motor current flowing through the motor winding **13** is at least substantially formed by the torque-forming current component I_q . Thus, when operating the machine **9** in the standard mode, the motor current is at least substantially free of a positive or negative field-forming current component I_d . In step S7, the motor winding is also energized in such a way that, due to the torque-forming current component I_q , it is achieved that the electric machine **9** generates the desired or specified desired torque $M_{sub.desired}$.

Claims

1. A method for operating an electric machine having a rotatably mounted rotor (**10**) and at least one motor winding (**13**), the method comprising: determining a temperature load of the machine (**9**), and applying an electric motor current to the motor winding (**13**) to generate a specified desired torque ($M_{sub.desired}$) according to the temperature load, wherein when the temperature load exceeds a specified load threshold, a field strengthening mode of the machine (**9**) is activated, wherein the motor winding (**13**) in the field-strengthening mode is energized in such a way that the motor current has a positive field-forming current component (I_d), and wherein when the temperature load falls below the load threshold, the motor winding (**13**) is energized in such a way that a torque-forming current component (I_q) of the motor current at least substantially matches the motor current.
2. The method according to claim 1, wherein the motor winding (**13**) is energized in the field-strengthening mode in such a way that the percent share of the positive field-forming current component (I_d) in the motor current is at least 10%, preferably at least 20%, particularly preferably at least 50%.
3. The method according to claim 1, wherein the motor winding (**13**) is energized in the field-strengthening mode in such a way that the positive field-forming current component (I_d) is the largest as possible percent share of the motor current allowing the machine (**9**) to generate the specified desired torque ($M_{sub.desired}$).
4. The method according to claim 1, wherein the temperature load of the rotor (**10**) is ascertained as the temperature load of the machine (**9**).
5. The method according to claim 1, wherein a threshold speed is specified for the rotor (**10**), and that the field-strengthening mode is activated only in the presence of an actual speed ($N_{sub.actual}$) of the rotor (**10**) that is below the threshold speed.
6. The method according to claim 1, wherein a threshold torque is specified for the desired torque ($M_{sub.desired}$) and that the field-strengthening mode is only activated in the presence of a desired torque ($M_{sub.desired}$) that is below the threshold torque.
7. The method according to claim 1, the method further comprising: determining an actual speed ($N_{sub.actual}$) of the rotor (**10**) is below a threshold speed of the rotor (**10**), determining a desired torque ($M_{sub.desired}$) is below a threshold torque for the desired torque ($M_{sub.desired}$), and activating the field strengthening mode of the machine (**9**) in response to the temperature load exceeding the specified load threshold, the actual speed ($N_{sub.actual}$) being below the threshold speed, and the desired torque ($M_{sub.desired}$) being below the threshold torque.
8. The method according to claim 1, wherein the temperature load of the machine (**9**) is a temperature of the rotor (**10**) based on a temperature of a stator (**23**) of the machine (**9**).
9. A device for operating an electric machine, wherein the machine (**9**) comprises a rotatably mounted rotor (**10**) and at least one motor winding (**13**), wherein the device (**19**) comprises a

control unit (20) configured to determine a temperature load of the machine (9), and apply an electric motor current to the motor winding (13) to generate a specified desired torque (N.sub.desired) according to the temperature load, wherein when the temperature load is determined that exceeds a specified load threshold, a field strengthening mode of the machine (9) is activated, wherein the motor winding (13) in the field-strengthening mode is energized in such a way that the motor current has a positive field-forming current component (I_d), wherein when the temperature load falls below the load threshold, the motor winding (13) is energized in such a way that a torque-forming current component (I_q) of the motor current at least substantially matches the motor current.

10. An electrical machine having a rotatably mounted rotor (10), at least one motor winding (13), and a device (19) according to claim 9.

11. A motor vehicle having an electric machine (9) according to claim 10.
